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Properties of Materials
Lecture 14

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Lecture 11: Packing of Atoms in Solids (2), Learning Objectives, Atomic Packing, Atomic Packing - bcc Structures, Atomic Packing Factor - bcc, Atomic Packing - fcc Structures, Atomic Packing Factor - fcc, Atomic Packing - hcp Structures, Atomic Packing Factor - hcp, Tips for Calculating APF for hcp, Concepts and Terminology (6)
Lecture 12: The Physical Basis of Young's Modulus, Learning Objectives, Moduli of Crystals, Rubbers and the Glass Transition Temperature, Composite, Composite Terminology, Fibre-Reinforced Composites, Rule of Mixtures (1)
Lecture 13: Rule of Mixtures (2), Task, Concepts and Terminology (7)
Exercise Seminar 3
Lecture 14: The Yield Strength, Tensile Strength, Hardness and Ductility, Case Studies, Learning Objectives, How do Crystalline Materials Fail (1), Length Scales, Continuum Plastic Flow, Shearing, Shearing of a Ductile Polycrystal, Plastic Instability, What is Creep, Where is Creep Important

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The yield strength, tensile strength, hardness and ductility

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Learning objectives

- Understand the concept of **length scales** with regards to different perspectives on material properties:
  - Atomistic
  - Continuum
- Learn to describe and derive the requirements for macroscopic **plastic flow** by considering the effects of shear stress on a microscopic level
- To describe the reasons why **necking** occurs in ductile materials and the origins of plastic instability
- To learn about the fundamentals of high temperature deformation or **creep** of materials

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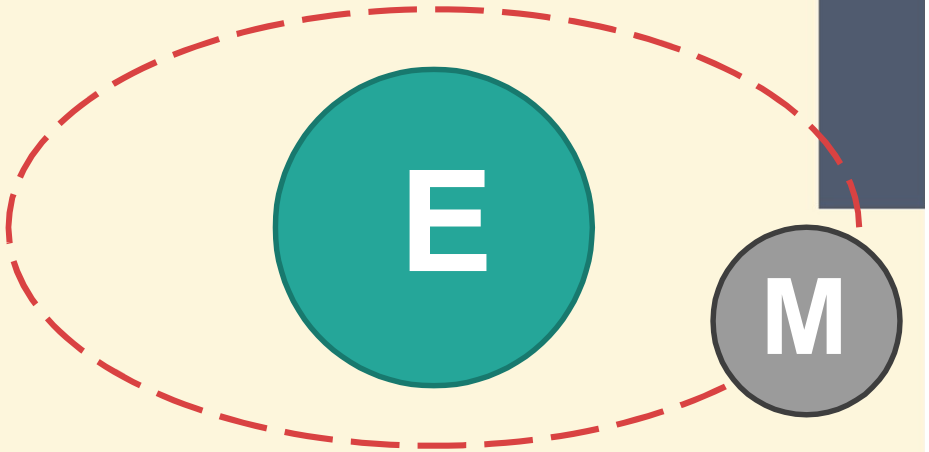
How do crystalline materials fail?

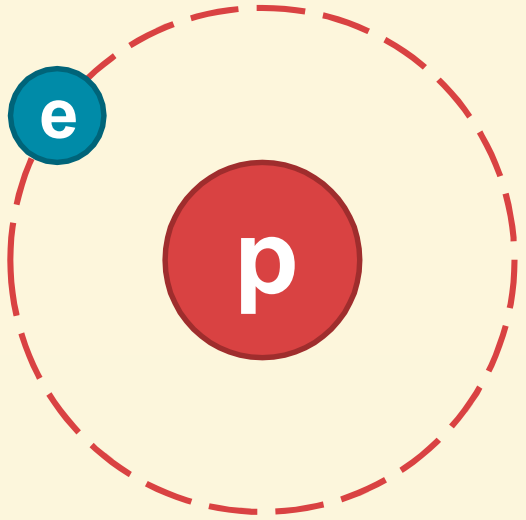
- **Plastic deformation**
  - Low temperature yielding
  - High temperature creep
- **Fracture**
  - Brittle fractures
  - Ductile fractures
- **Fatigue**
  - Slow crack growth leading to fracture
- **Oxidation** and **Corrosion**
  - Loss of material due to reactions with O_2, and $O_2 + H_2O$
- **Wear**
  - All of the above, with relative motion between parts!

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Length scales

- Range of distances (of a few orders of magnitude) over which distinct theories apply
- In materials:
  - **Atomistic** is approximately 10^{-10} m
  - **Continuum** is $>10^{-7}$ m…maybe…
- The 'rules' which apply **depend on the scale** being considered





Question – Which of the following systems is more 'closely packed': The Earth and The Moon? A hydrogen atom (1p + 1e)?

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Continuum plastic flow

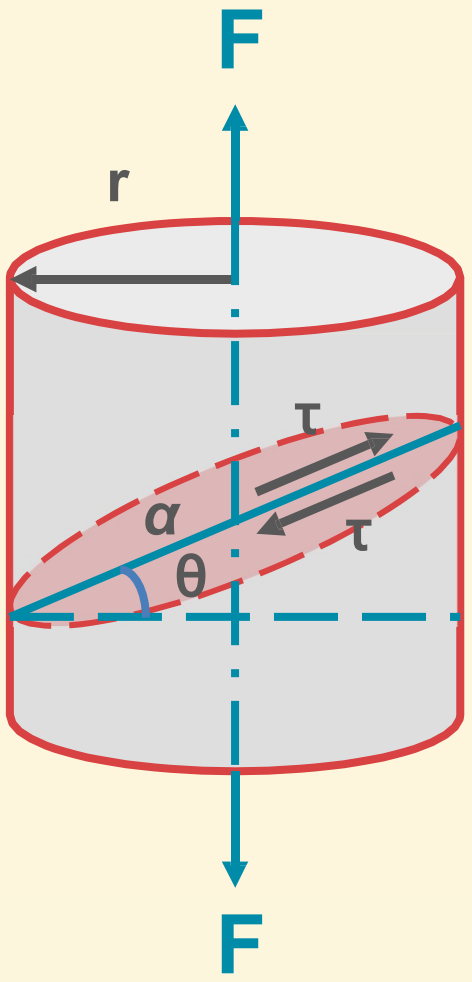
- Continuum – non-atomistic; **macroscopic** behaviour

- Plastic flow occurs by **shearing**:
  - Dislocations move when $\tau_{\text{slip plane}} > \tau_{y\perp}$ of a single crystal
  - Averaged over all grain orientations and slip planes, it can be related to the yield strength of a polycrystalline material: $\sigma_y \approx 3\tau_{y\perp}$

- However, need **shear yield strength** of a polycrystalline material – i.e. shear resistance averaged over all grain orientations and slip planes

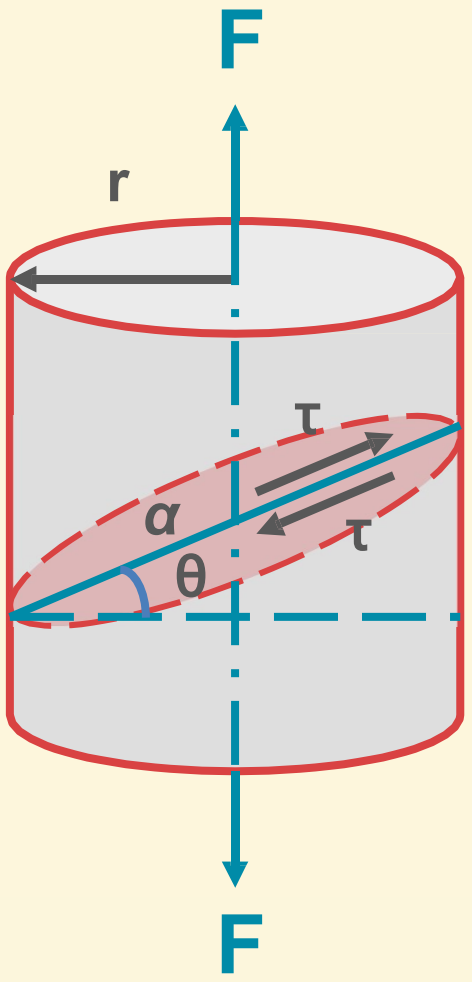
Shearing (1)

- Slip planes within a loaded structure can often be identified or guessed:
  - **Preferred slip planes** (i.e. close-packed planes)
- Failure load can therefore be calculated by requiring **stress exceeds shear yield strength**:
  - $\tau_y = \frac{\sigma_y}{2}$
- Proof…?



Shearing (2)

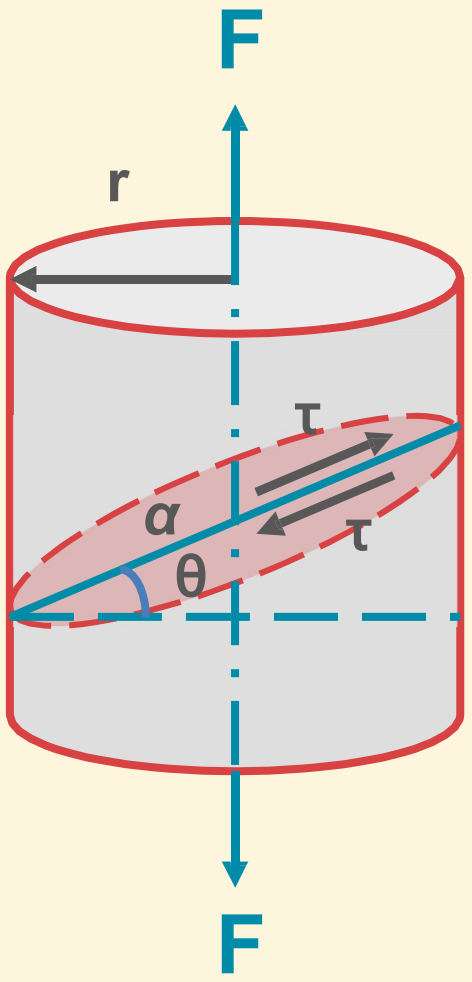
- Tensile force 90° to plane:
  - $F_T = F \cos\theta$
- Shear force parallel to plane:
  - $F_S = F \sin\theta$
- Area of tensile plane:
  - $A_T = \pi r^2$
- Area of shear plane (ellipse):
  - $A_S = \pi a r$



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Shearing (3)

- $F_T = F\ \cos\theta$; $F_S = F\ \sin\theta$
  $A_T = \pi r^2$; $A_S = \pi a r$
- Relate *a* to *r*:
  - $a = \frac{r}{\cos\theta}$
  - $\Rightarrow A_S = \frac{A_T}{\cos\theta}$
- Shear stress:
  - $\tau = \frac{F_S}{A_S} = \frac{F\ \sin\theta\ \cos\theta}{A_T}$
  - $\Rightarrow \tau = \sigma \sin\theta \cos\theta = \sigma \frac{\sin 2\theta}{2}$

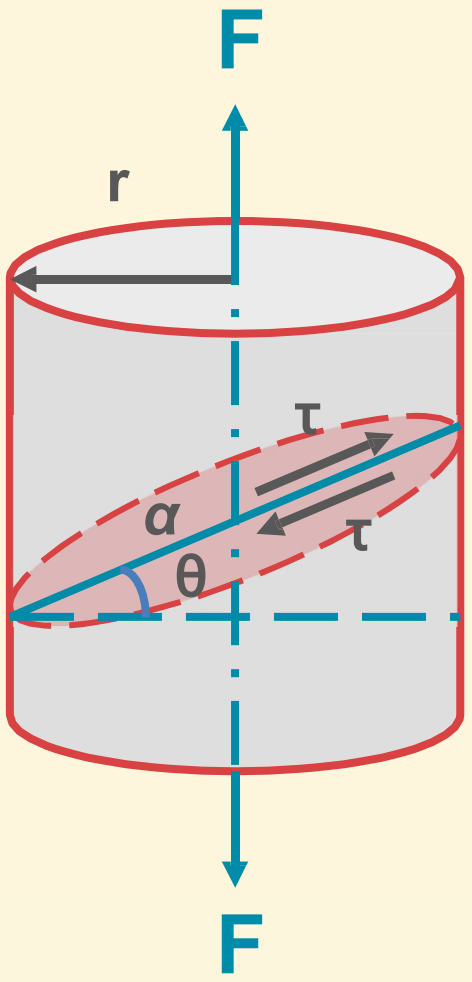


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Shearing (4)

- Shear stress:
  - $\tau = \sigma \sin\theta \cos\theta = \sigma \frac{\sin 2\theta}{2}$

- Maximum when θ =45°:
  - $\tau_{max} = \frac{\sigma}{2}$
  - $\therefore \boxed{\tau_y = \frac{\sigma_y}{2}}$



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Shearing of a ductile polycrystal (1)

- Shear is related to slip
- In a single crystal (A), slip occurs on **nearest plane to 45°**
- In a polycrystal (B), slip occurs in multiple grains with different orientations
- The **average slip direction is 45°**

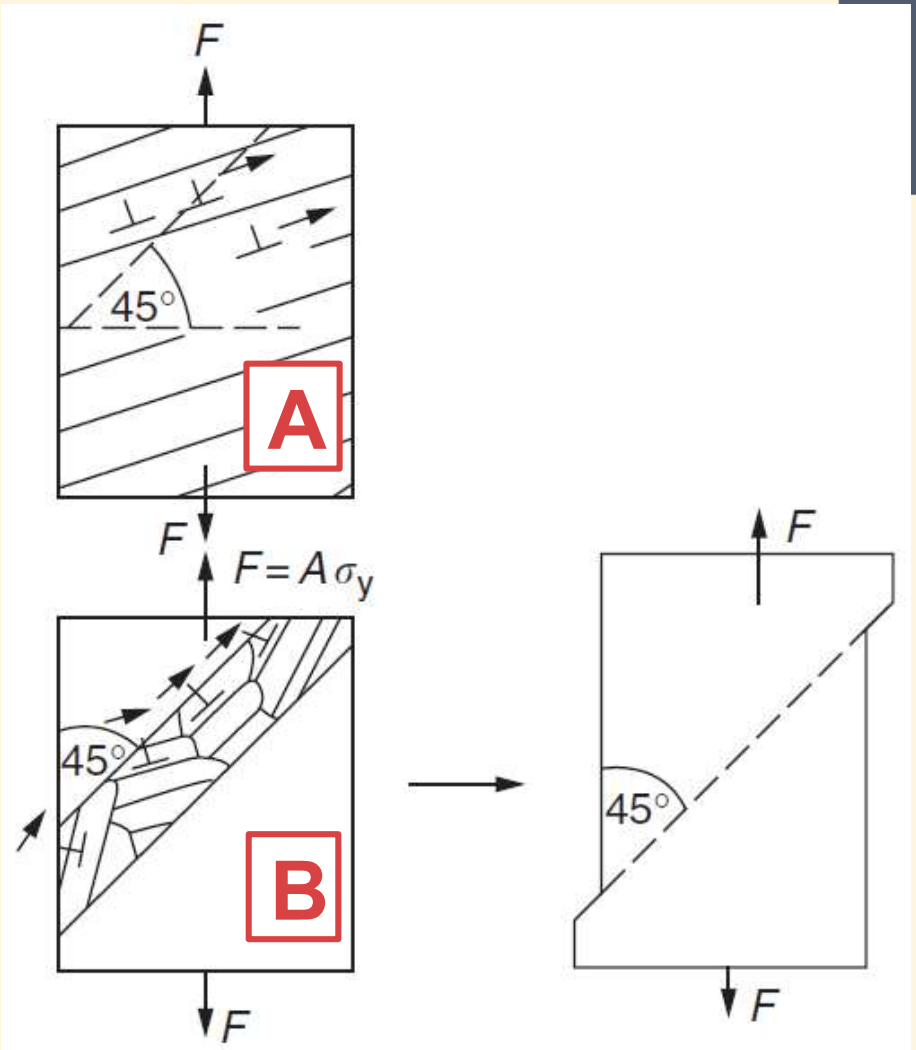


Image from: Ashby, M.F., Jones, D.R.H., (2012), *Engineering Materials 1*



‘Cup and cone’ fracture



Shear fracture

Image from: Kabir, I.R. and Islam, M.A., (2014), *Hardened Case Properties and Tensile Behaviours of TMT Steel Bars*, American Journal of Mechanical Engineering, Vol. 2, No. 1, p. 8-14, doi: 10.12691/ajme-2-1-2, Science and Education Publishing.

Shearing of a ductile polycrystal (2)

- Results in characteristic fracture surfaces
- Whether ‘cup and cone’ or shear depends on whether material exhibits brittle behaviour in centre

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Plastic instability (1)

- Why do ductile materials experience **necking**?

- When yield strength reached, one of two things can occur:
  - **Strength rises** to sustain stress (i.e. strain hardening)
  - **Plastic flow** (i.e. necking)
- **Critical condition** for necking:
  - $A\sigma = F =$ constant

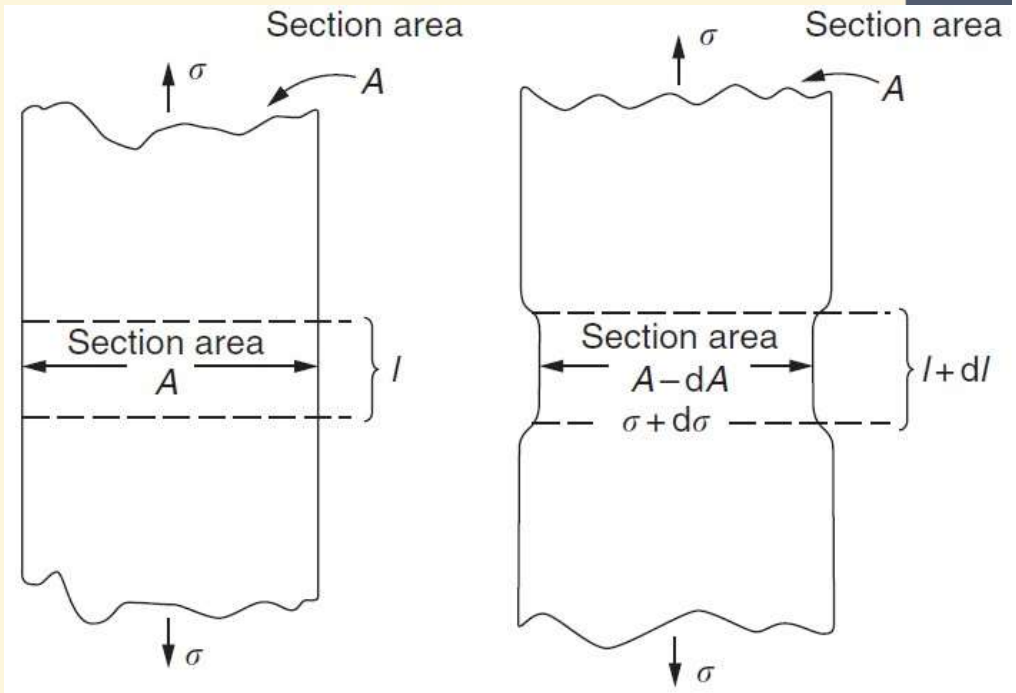


Image from: Ashby, M.F., Jones, D.R.H., (2012), *Engineering Materials 1*

Plastic instability (2)

- **Critical condition** for necking:
  - $A\sigma = F =$ constant
- Considering incremental changes:
  - $A\mathrm{d}\sigma + \sigma\mathrm{d}A = 0$
  - $\Rightarrow \frac{\mathrm{d}\sigma}{\sigma} = -\frac{\mathrm{d}A}{A}$
- Assume **volume is conserved**:
  - $-\frac{\mathrm{d}A}{A} = \frac{\mathrm{d}l}{l} = \mathrm{d}\varepsilon$
  - $\Rightarrow \frac{\mathrm{d}\sigma}{\mathrm{d}\varepsilon} = \sigma$

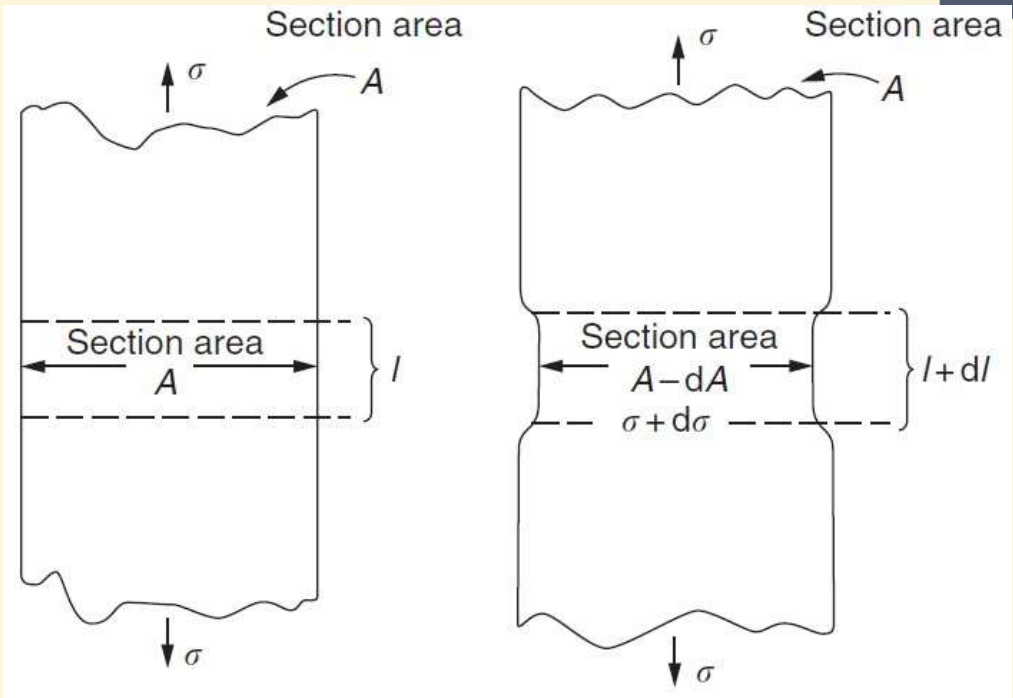


Image from: Ashby, M.F., Jones, D.R.H., (2012), *Engineering Materials 1*

Plastic instability (3)

- Assume **volume is conserved**:
  - $-\frac{dA}{A} = \frac{dl}{l} = d\varepsilon$
  - $\Rightarrow \frac{d\sigma}{d\varepsilon} = \sigma$
- This is **true stress and strain** – working in nominal values:
  - $\sigma_t = \sigma_n(1+\varepsilon_n)$
  - $\varepsilon_t = \ln(1+\varepsilon_n)$

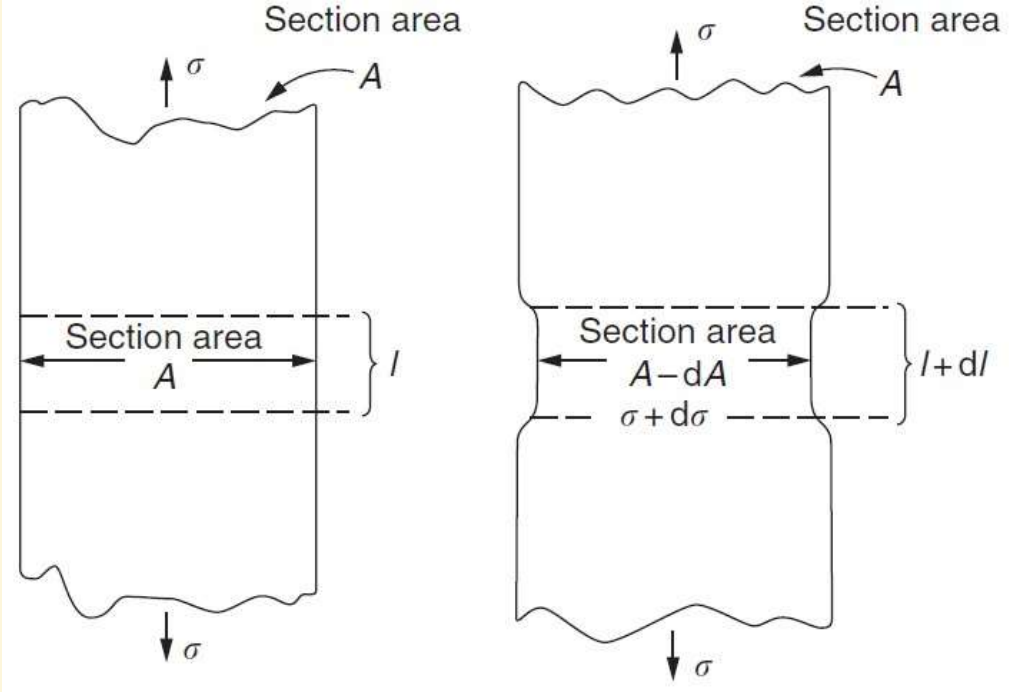


Image from: Ashby, M.F., Jones, D.R.H., (2012), *Engineering Materials 1*

- Differentiating these and substituting in results in:
  - $\frac{d\sigma_n}{d\varepsilon_n} = 0$

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Plastic instability (3)

$$\frac{d\sigma_t}{d\varepsilon_t} = \sigma_t$$

$$\varepsilon_t = \ln(1+\varepsilon_n) \Rightarrow d\varepsilon_t = \frac{d\varepsilon_n}{1+\varepsilon_n}$$

$$\sigma_t = \sigma_n(1+\varepsilon_n) \Rightarrow d\sigma_t = (1+\varepsilon_n)d\sigma_n + \sigma_n d\varepsilon_n \Rightarrow$$

$$\sigma_t\, d\varepsilon_t = (1+\varepsilon_n)d\sigma_n + \sigma_n d\varepsilon_n \Rightarrow$$

$$\sigma_n(1+\varepsilon_n)\frac{d\varepsilon_n}{(1+\varepsilon_n)} = (1+\varepsilon_n)d\sigma_n + \sigma_n d\varepsilon_n \Rightarrow$$

$$\sigma_n\, d\varepsilon_n = (1+\varepsilon_n)\, d\sigma_n + \sigma_n d\varepsilon_n \Rightarrow$$

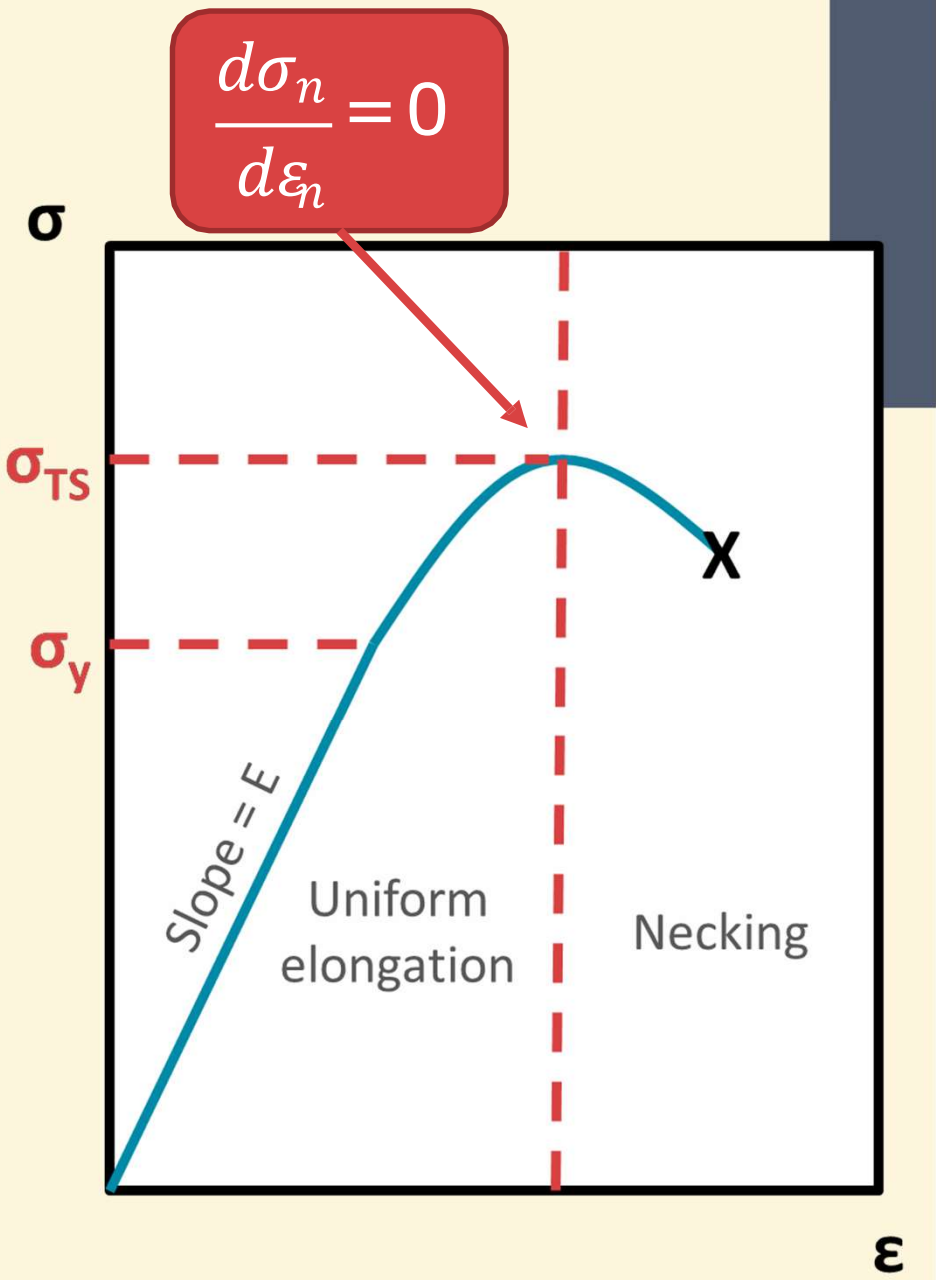
$$\sigma_n = (1+\varepsilon_n)\frac{d\sigma_n}{d\varepsilon_n} + \sigma_n \Rightarrow$$

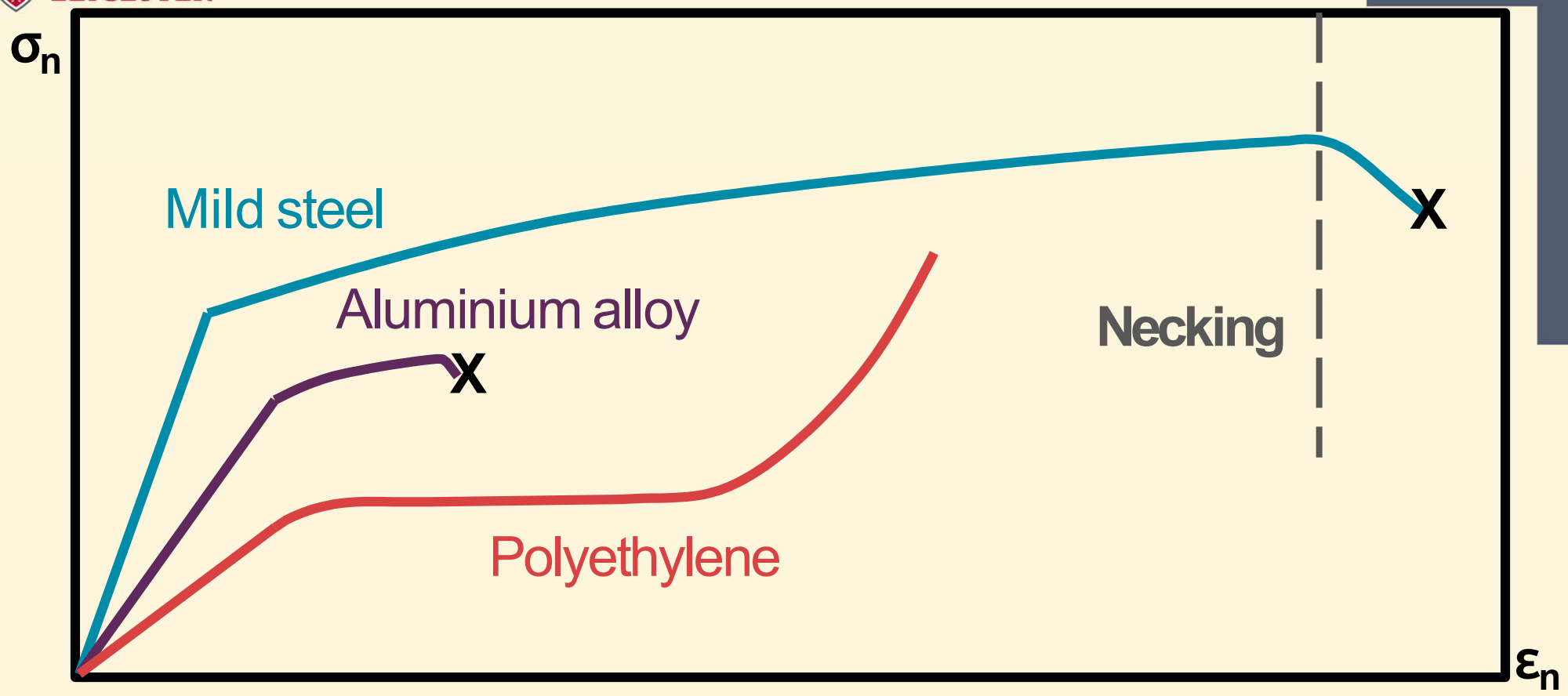
$$0 = (1+\varepsilon_n)\frac{d\sigma_n}{d\varepsilon_n}$$

Because $1+\varepsilon_n > 0$, therefore $\frac{d\sigma_n}{d\varepsilon_n} = 0$.

Plastic instability (4)

- Condition for necking to start:
  - $\frac{d\sigma_n}{d\varepsilon_n} = 0$
- Physically, the onset of non-linearity between nominal stress and strain is **work hardening**
- Different materials have different capacities for work hardening…





Plastic instability (5)

- Mild steel is able to **flow significantly** before necking
- Aluminium **work hardens rapidly** and therefore necks easily
- Polyethylene **necks easily**, but in a **stable** fashion

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What is creep?

- Slow, irreversible deformation:
  - Occurs at stress levels **below yield strength**
  - Occurs at **high temperatures** ($> 0.3T_m$ for metals; $> 0.4T_m$ for ceramics)

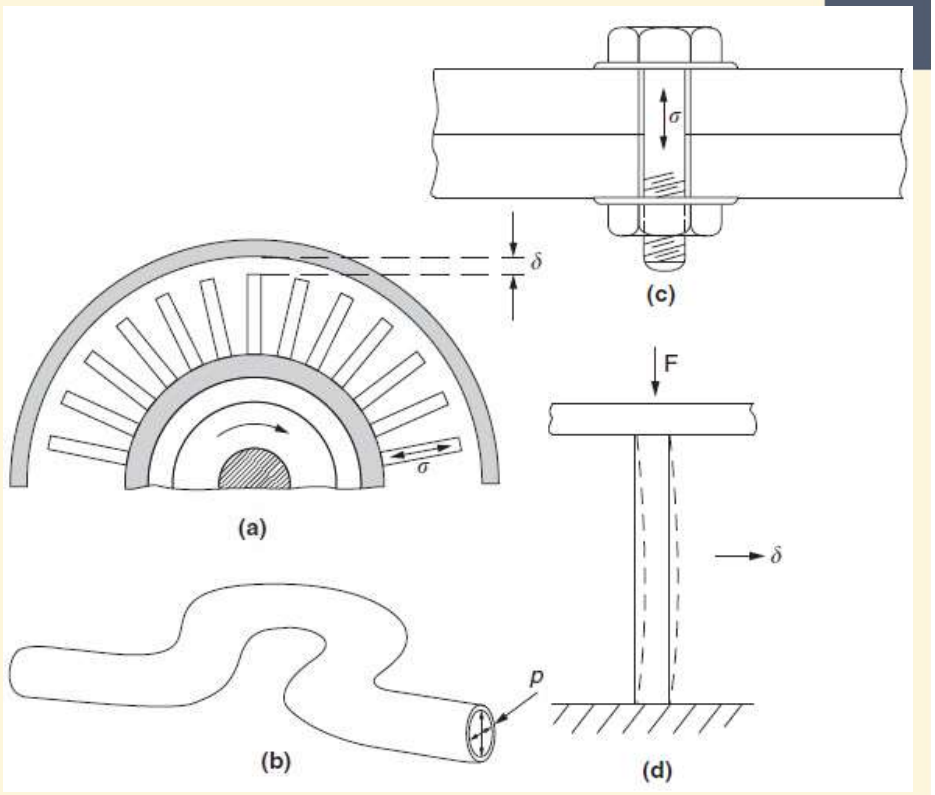
- 'Accumulative' process:
  - Deformation is **time-dependent**

- Extent and rate of creep depend on:
  - Material properties and structure
  - Exposure time
  - Temperature
  - Applied load

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Where is creep important?

- **Displacement**-limited designs:
  - Precise dimensions / tolerances must be maintained – e.g. turbine blades
- **Rupture**-limited designs:
  - Fracture must be avoided – e.g. high-pressure pipes
- **Stress relaxation**-limited designs:
  - Reduction of tension with time – e.g. pre-tension bolts
- **Buckling**-limited designs:
  - Compressive loading of slender columns – e.g. structural steel



Images from: Ashby, M.F., Jones, D.R.H., (2012), *Engineering Materials 1*